

Lab 3 – Architectural Pattern Selection

Topic: Second-Hand Product Escrow Marketplace

SRN: PES1UG24AM427 | Name: Dhanush R

Architectural Justification

We chose **Microservices Architecture** for the Second-Hand Product Escrow Marketplace.

The main reason for choosing this architecture is that the system has different functions such as **order management, payment and escrow, product inspection, and dispute management**. These functions can be separated into different services, making the system easier to manage and update. For example, changes to the inspection service would not require changes to the payment service.

Another reason is that the different parts of the marketplace may have different workloads. Payment and escrow operations may need more resources when many users are making transactions, while the inspection or dispute services may have less activity. With microservices, each service can be scaled separately when required.

From a **security** point of view, keeping the payment and escrow functionality as a separate service helps protect sensitive payment operations. Access to this service can be restricted so that only the required components can communicate with it.

For **performance**, individual services can be scaled independently. For example, if there are many orders being placed, the Order Manager can be scaled without having to scale the entire application. This can help the system handle more users efficiently.

Overall, Microservices Architecture is suitable for this marketplace because it separates the major functions of the system while making the system easier to scale, maintain, and secure.

